

Subject: Physical Geography | NCERT Class 7 Ch.6 & Class 11 Ch.14 | UPSC & State PSC Core

30-Second Snapshot

- Natural vegetation is dictated primarily by **temperature and moisture**, and secondarily by **slope and soil depth**.
- Vegetation zones shift with **latitude** (Equator to Poles) in parallel with **altitude** (Foothills to Mountain Peaks).
- Biodiversity exists across 3 levels: **Genetic**, **Species**, and **Ecosystem**. 12 tropical mega-diversity countries account for the majority of global species richness.

World Forest & Vegetation Matrix

Forest / Biome Type	Global Distribution	Climatic Adaptation & Mechanism	Key Tree Species	Typical Fauna
Tropical Evergreen	Equatorial belt (Amazon basin, Congo, SE Asia)	Heavy rainfall year-round; dense multi-layered canopy; no distinct leaf-fall season	Rosewood, Ebony, Mahogany (Hardwood)	Anaconda, jaguars, canopy primates
Tropical Deciduous	Monsoon zones (India, N. Australia, Central America)	Pronounced dry season; trees shed leaves for 6–8 weeks to prevent transpiration	Teak, Sal, Shisham, Neem, Sandalwood	Bengal tiger, Asian elephant, langur
Temperate Evergreen	Warm mid-latitude eastern continental margins (SE USA, S China, SE Brazil)	Moderate rainfall, maritime influence	Oak, Pine, Eucalyptus	Deer, diverse bird life
Temperate Deciduous	Cool mid-latitudes (NE USA, Western Europe, NE Asia, Chile)	Cold winters; shed leaves during autumn/winter season	Oak, Ash, Beech	Red fox, wolves, pheasants, monals
Mediterranean	Western continental margins (30°–45° N & S): Mediterranean Basin, California, Central Chile, SW Australia	Hot dry summers & mild wet winters ; thick bark, waxy leaves, deep roots	Olive, Cork Oak, Citrus fruits, Fig, Vineyards	Limited native fauna; agro-cultural landscapes
Coniferous (Taiga)	Sub-polar belt (50°–70° N); high mountain slopes	Cone-shaped canopy to shed snow; needle-like leaves; softwood timber for paper pulp	Chir, Pine, Cedar, Silver Fir, Spruce	Lynx, mink, sable, silver fox
Tundra	Arctic shores & mountain alpine zones beyond tree line	Permafrost soil; extremely short summer growing season	Mosses, Lichens, Dwarf willow	Musk-ox, Arctic hare, Reindeer, Polar bear

Global Grasslands Classification

- Tropical Grasslands (Tall, coarse, 3–4m tall):**
 - Savanna:** East Africa
 - Campos:** Brazilian Highlands
 - Llanos:** Venezuela & Colombia
- Temperate Grasslands (Short, highly nutritious, world breadbaskets):**
 - Prairie:** North America
 - Pampas:** Argentina
 - Steppes:** Central Eurasia
 - Veld:** South Africa
 - Downs:** Australia

Visual Mental Model: Altitudinal vs. Latitudinal Parallels

- High Altitude / Latitude (Poles / Peaks):**
 - Tundra:** Mosses, lichens, permafrost
 - Taiga:** Coniferous needle-leaf softwoods (Pine, Fir)
 - Temperate:** Broad-leaved deciduous (Oak, Beech)
 - Subtropical:** Pine, mixed forests
 - Tropical:** Dense broadleaf evergreen / monsoon deciduous
- Low Altitude / Latitude (Sea Level / Equator)**

⚠ Common UPSC Exam Traps

- **Trap 1 (Mediterranean Climate):** Aspirants often assume rain falls in summer. **Fact:** Mediterranean zones get rain in **winter** under Westerlies; summers are completely **dry**.
- **Trap 2 (Taiga vs Tundra):** Taiga supports tall softwood forests (commercial timber/paper pulp); Tundra is **treeless** due to frozen subsoil (permafrost).
- **Trap 3 (Wood Type):** Tropical rainforests produce **hardwood** (difficult to extract commercially due to species heterogeneity); Coniferous forests produce **softwood** in pure single-species stands (ideal for commercial logging).

🎯 High-Yield Facts & Treaties

Parameter	Core Detail	Institutional Anchor
12 Megadiversity Nations	Mexico, Colombia, Ecuador, Peru, Brazil, DRC, Madagascar, China, India, Malaysia, Indonesia, Australia	UNEP-WCMC Classification
India Legal Framework	Wildlife (Protection) Act, 1972	Establishes National Parks & Sanctuaries
Earth Summit 1992	Rio de Janeiro Convention on Biological Diversity (CBD)	Legally binding global framework
IUCN Red List	Classifies threatened status: Endangered, Vulnerable, Rare	IUCN Gland, Switzerland

📝 Practice Challenge (Prelims-Standard)

Q1. Consider the following statements regarding world natural vegetation:

1. In Mediterranean regions, trees have thick bark and wax-coated leaves to restrict transpiration during hot, dry summers.
2. Taiga vegetation is predominantly composed of hardwood trees prized for luxury furniture.
3. Campos and Llanos are regional examples of temperate grasslands.

Which of the statements given above is/are correct?

- (A) 1 only
- (B) 1 and 2 only
- (C) 2 and 3 only
- (D) 1, 2 and 3

Answer: (A) 1 only

Explanation: Statement 2 is incorrect because Taiga consists of *softwood* trees (pine, cedar) used for pulp and paper. Statement 3 is incorrect because Campos and Llanos are *tropical* grasslands, while Pampas and Prairies are temperate.

Q2. With reference to the Earth's biodiversity, consider the following:

1. Species diversity consistently increases as one moves from the equator toward the polar regions.
2. More diverse ecosystems exhibit greater resilience and stability against environmental disturbances.

Which of the statements given above is/are correct?

- (A) 1 only
- (B) 2 only
- (C) Both 1 and 2
- (D) Neither 1 nor 2

Answer: (B) 2 only

Explanation: Statement 1 is incorrect because biodiversity is richest in the tropics and consistently *decreases* toward the poles due to solar energy gradients. Statement 2 is correct.